

MD. Delwar Husen

Machine Learning Engineer

mohammaddelwar2004@gmail.com | +880 1609-284107 | LinkedIn | GitHub | Portfolio | Dhaka, Bangladesh

SUMMARY

Machine Learning Engineer-focused undergraduate with experience in Machine Learning, Deep Learning, Computer Vision, and Medical AI. Develops, trains, validates, and evaluates ML systems for real-world AI solutions and research-driven engineering.

TECHNICAL SKILLS

Programming: Python

ML/DL: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch

Tools: Git, GitHub, Jupyter Notebook, Google Colab, Kaggle

EXPERIENCE

Machine Learning Engineering Intern — FlyRank AI

Jul 2026 – Aug 2026

Remote

- Developed end-to-end ML workflows on anonymized search-performance data, covering data analysis, feature engineering, baseline modeling, validation, and reproducible experimentation.
- Built a Random Forest-based editorial opportunity prediction framework using search visibility, CTR, search position, engagement, and traffic signals; achieved **98.68% accuracy, 95.78% F1, and 99.75% recall**, outperforming a rule-based baseline under the evaluated setup.
- Implemented grouped client-level validation and leakage audits, then transformed model predictions into a ranked content action playbook with interpretable reason codes and deployed the findings as a public research paper.

Deployed Research Paper: delwar2004.github.io/flyrank-ml-internship

RESEARCH EXPERIENCE

Medical AI Research — 2 ongoing research projects focused on machine learning applications in healthcare.

SELECTED PROJECTS

Suspicious Behavior Detection — YOLOv10n vs RT-DETR Benchmark

Computer Vision

- Benchmarked YOLOv10n and RT-DETR for 4-class suspicious behavior object detection on a custom dataset of **1,929 images** (1,688 train, 161 validation, 80 test) under identical experimental conditions.
- RT-DETR achieved **0.9455 mAP@50, 0.9093 precision, 0.9615 recall, and 0.9340 F1-score**, while YOLOv10n achieved **0.9255 mAP@50, 0.9189 precision, 0.9130 recall, and 0.9159 F1-score**.

Technologies: Python, PyTorch, Ultralytics, YOLOv10n, RT-DETR, AdamW, AMP

Real-Time Violence Classification using YOLO11n

Computer Vision

- Developed a lightweight binary violence classification pipeline using YOLO11n Classification on the Hockey Fights Dataset, covering preprocessing, training, validation, evaluation, and confusion matrix analysis.
- Achieved **98.50% accuracy, 98.99% precision, 98.00% recall, and 98.49% F1-score** on 400 validation images, with **394 correct and 6 incorrect predictions**, including only 2 false negatives and 4 false positives.

Technologies: Python, PyTorch, Ultralytics, YOLO11n, OpenCV, NumPy, Matplotlib, Scikit-learn

Movie Recommendation System — Collaborative Filtering & Matrix Factorization

Recommender Systems

- Developed an end-to-end recommendation pipeline on MovieLens Latest-Small with **610 users, 9,742 movies, and 100,836 ratings**, implementing popularity-based, user-based and item-based collaborative filtering, and SVD matrix factorization.
- User-Based CF achieved the highest **Precision@10 of 0.3672**, while SVD achieved the best **RMSE of 0.8807**; Item-Based CF achieved **0.2164 Precision@10** and **0.9800 RMSE**.

Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Surprise, SciPy

Loan Approval Prediction — Machine Learning Classification

Machine Learning

- Developed an end-to-end loan approval classification pipeline using applicant demographic and financial features, covering data quality assessment, EDA, preprocessing, model development, and comparative evaluation.
- Compared Logistic Regression and Decision Tree with and without SMOTE; Decision Tree achieved **97.42% accuracy, 98.66% precision, 97.18% recall, and 97.91% F1-score**, outperforming all evaluated models.

Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Imbalanced-learn

EDUCATION

Bangladesh University of Professionals (BUP)

B.Sc. in Information and Communication Engineering

2024 – Present

6th Semester | CGPA: 3.22/4.00

Expected Graduation: 2027

EXTRACURRICULAR ACTIVITIES

Assistant Tech Master, BUP InfoTech Club - Support technical workshops, member activities, and club initiatives.

CERTIFICATIONS

Data Science & Machine Learning with Python — Ostad

Artificial Intelligence & Machine Learning Fundamentals — Grameenphone Academy

Certificate

Certificate

LANGUAGES

Bangla: Native

English: Professional Proficiency